

Expressions, Formulae and Brackets — Paper B

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper.
- When you substitute a value, **write it in brackets**: $3(-2)$, not $3-2$. Most lost marks on this paper are dropped signs.
- Only **like terms** can be added — terms with exactly the same letter part. $3x + 2$ is already simplified.
- “Factorise completely” means take out the **highest** common factor. Check by looking inside your bracket: if the numbers there still share a factor, you stopped early.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – SUBSTITUTING AND SIMPLIFYING

1. Complete the table. Each empty box is worth one mark. Write the number in brackets as you substitute it. [6]

Expression	Value when $x = 3$	Value when $x = -2$
$2x + 5$		
x^2		
$3x - 4$		

The middle row is the one that catches people: squaring a negative gives a positive.

2. Simplify:

(a) $6y + 3 - 2y - 8$ [1]

(b) $7ab - 3ab$ [1]

3. Using n for the number, write an expression for each of these.

(a) three times the sum of the number and five [1]

(b) the number squared, then doubled

[1]

SECTION 2 – BRACKETS, BOTH DIRECTIONS

4. Expand and simplify:

(a) $2(x + 3) + 3(x + 1)$

[2]

(b) $4(2x - 1) + 3(x + 5)$

[2]

5. Factorise completely:

(a) $12x + 8$

[1]

(b) $5y^2 + 10y$

[1]

(c) $9x - 12$

[1]

(d) $4a^2 + 6a$

[1]

6. Expand:

(a) $x(x + 7)$

[1]

(b) $2x(3x - 1)$ [1]

(c) $(x + 2)(x + 5)$ [1]

(d) $3(2x + 1)$ [1]

SECTION 3 – USING IT

7. A triangle has sides of length x cm, $(x + 3)$ cm and $(2x - 1)$ cm.

(a) Write an expression for the perimeter, fully simplified. [2]

(b) Find the perimeter when $x = 5$, then check your answer by working out the three side lengths separately and adding them. [2]

8. A pen costs p rupees and a notebook costs n rupees.

(a) Write an expression for the cost of 4 pens and 3 notebooks. [1]

(b) Each of 5 students buys 4 pens and 3 notebooks. Write an expression for the total cost, with the bracket expanded. [1]

(c) Find that total when $p = 12$ and $n = 45$.

[2]

End of paper. Check every answer has a unit or a form the question actually asked for.